How AI Sees KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

Word bank: 0 3 255 blue

1. Color by number. Color every square with a **1**. Leave every **0** white.

0	0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	1	1	0	0
0	1	1	1	1	0	0	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	0	0
0	0	0	1	1	1	1	1	1	0	0	0
0	0	0	0	1	1	1	1	0	0	0	0
0	0	0	0	0	1	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0

What is the picture? a heart

2. Pixels are numbers.

In a black-and-white picture, 0 is black and 255 is white.

In a color picture, each pixel is 3 numbers: red, green, and blue.

Word bank: corners edges neuron objects weights

You can use a word more than once.

3. A filter is a neuron.

A filter is a neuron with 9 inputs. Its 9 numbers are its weights.

0	0	255
0	0	255
0	0	255

×

−1	0	+1
−1	0	+1
−1	0	+1

Total = 765 Does it fire? yes

If all 9 picture numbers are 255, the total is 0. Is there an edge? no

4. Layers.

Layer 1 finds edges. Layer 2 looks at layer 1's maps, not the picture. It finds corners.

In the pictures from a real network, early layers find edges. Later layers find parts and whole objects.

Who chose the real network's filters? Nobody. Training found them.

5. Watch the layers. Count the corners yourself first. Then check the network.

Shape	My count	The network's count
square	4	4
letter L	6	6
letter T	8	8
plus	12	12
circle	0	4

Why did the network find corners on the circle? Its pixels make tiny steps. Each step is a tiny corner.

6. Talk about it. What could a self-driving car miss?

A self-driving car could make a mistake when Answers vary. It sees something new.

Go Further (Day 2). Find the edge by hand. Here is a tiny picture (0 = dark, 9 = bright) and a filter.

0	0	0	9	9	9
0	0	0	9	9	9
0	0	0	9	9	9

Filter:

−1	0	1
−1	0	1
−1	0	1

Put the filter over 3 columns of the picture. Multiply each pair of numbers, then add all 9.

Filter over columns	Math	Answer
1, 2, 3	each row: $-0+0+0=0$	0
2, 3, 4	each row: $-0+0+9=9$, so $9+9+9$	27
3, 4, 5	each row: $-0+0+9=9$	27
4, 5, 6	each row: $-9+0+9=0$	0

Where are the big answers? Where the picture changes from 0 to 9: the edge.